Encode London, 23 to 25 October 2026 · Anusha, Andrii and Nastia · github.com/anushaisawesome/unseen-steps · unseen-steps-studio.vercel.app

Unseen Steps: audio description for every studio's show

Show notes prepared from a rehearsal by Gemini 4 Argon, delivered live by Gemini Live in the gaps in the music, and checked against the dancer's measured movement.

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1. The problem

Dance is a visual art. For the more than two million people in the UK living with sight loss (RNIB), a dance show is mostly music. Professional audio description of dance exists, but it is expensive and goes to large venues for a performance or two per run. The shows put on by small studios, schools and competitions, where most dancers actually perform, rarely get it.

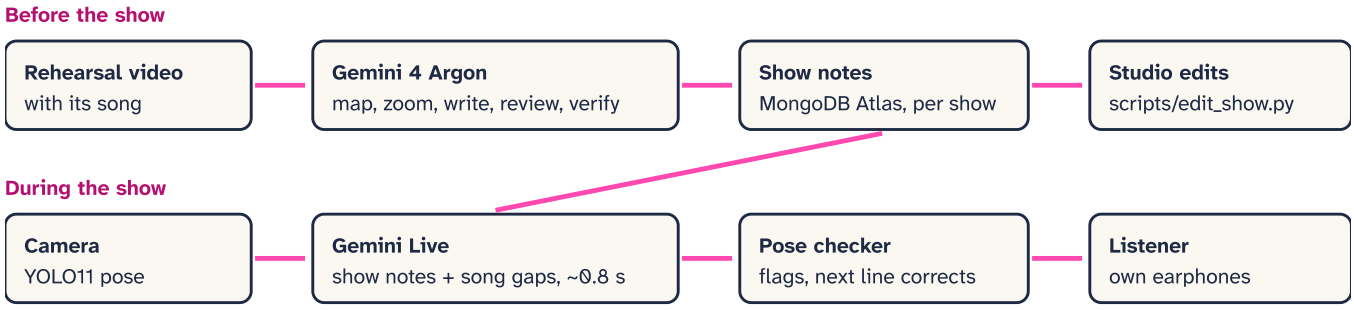
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2. Who it is for

Users: the blind and partially sighted people in the audience of a studio show, listening on their own phone and earphones. **Operators:** the studio, or a describer working with it. The project started with Anusha's robotics tutor, Gerard Canal (KCL), who dances ballroom with blind dancers; testing with his dancers is the next step. Andrii brought five years of ballroom dancing in Ukraine; Anusha brought the accessibility problem, and a dancer's eye from training in commercial, contemporary and ballet as a child and in street styles today, which informed her feedback on the narration across styles.

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3. How it works



Before the show: Gemini 4 Argon prepares show notes

- **Map** (whole video and music, 5 frames a second): style with visible evidence, sections with formations, group structure and techniques, the piece's signature, a timeline of texture and energy changes, motifs and key moments, as structured JSON.
- **Zoom**: each section re-watched at 8 frames a second, second by second.
- **Write**: the script is written a stretch of up to 20 s at a time (8 fps) with the whole map in context, in the style's own vocabulary explained for newcomers, one line per musical phrase plus callouts on hits.
- **Review**: a second Argon pass reads the script as a dancer would hear it and fixes, adds or drops lines.
- **Verify**: every line is checked against its exact moment at 10 frames a second; it corrects only what it can clearly see is wrong.
- **Store**: notes, map, gaps in the song and the rehearsal script go to MongoDB Atlas per show; the studio reads and edits them with `scripts/edit_show.py`.

During the show: Gemini Live narrates what actually happens

YOLO11 pose follows the dancer and measures jumps, turns and their direction, facing, footwork, floor work and turnout. One Gemini Live session holds the show notes and sees the camera at 2 frames a second. At each gap in the song it is prompted with the measurements (as a check, not a script) and the rehearsal line for that moment, and describes what the dancer does now, including anything different from rehearsal. It names a figure only if the routine contains it. A checker compares each spoken claim with the pose; a flagged line is reported in the next prompt and the next line corrects it.

Models

Model	Job
gemini-4-argon	Performance map, section zoom, windowed writing, second-describer review, per-line visual check, style check, final cut; running notes live
gemini-3.8-live	Live narration from the camera and the show notes (audio out)
gemini-3.8-flash-tts	Voice for recorded narration; every clip transcribed and checked against its caption
gemini-3.8-flash	Caption check, translation, regression judging
gemini-embedding-2	3072-dimension glossary vectors for Atlas Vector Search
deep-research-preview-04-2026	Sourced waltz glossary drafts, vetted by dancers before use
gemini-3-pro-image	Logo and site illustration
YOLO11 pose, librosa	Body tracking and measurements; beats, phrases and gaps in the music

MongoDB Atlas

Three Atlas features run in the product. **Hybrid Search** matches described movement to the dancer-vetted glossary by fusing Atlas Vector Search and an Atlas Search text index with reciprocal rank fusion (vetted terms only). **Atlas Search** finds a show's music map by name with fuzzy matching. Each show's notes, gaps, map and rehearsal script are stored per show and loaded live.

4. Verification loops and tracking safeguards

Every line is grounded more than once: the prepared script is checked against the video at each line's moment; every recorded voice clip must say exactly its caption or it is re-recorded or dropped; live, each claim is compared with the measured pose. When body tracking is unsure, the rule throughout is to measure nothing and say less (from `docs/tracking-safeguards.md`):

Layer	What it does
Pose confidence	Keypoints under 0.3 left out of measurements; legs and turnout only when hips, knees and ankles are clearly visible (over 0.5).
Coverage per phrase	Tracked in under 30% of a phrase: nothing measured, body claims rejected, plain wording allowed.
Unmeasurable is unchecked	Rotation in profile, jumps or turns in a group: the check is skipped rather than guessed.
Crowded footage	A marked couple is followed; if lost it goes silent and only re-acquires when pose and clothing both match.
Gemini sees the camera	Pose is sent as a check only, so a bad reading doesn't become the narration; a flagged line is corrected next.
People can see it	The skeleton shows who is tracked; a contact sheet shows who was followed before rendering.

5. Taste: a dancer's feedback loop

The narration was tuned by a dancer (Anusha) reviewing renders across styles. Her reviews live in `feedback/*.md` in her own words and are read by Argon on every run, so a fix applies to every style rather than one video. Examples from the commit history:

- **Frame rate.** Gemini reads video at 1 frame a second by default, too slow to see sharp versus fluid movement or arm detail. Raising every read to 5 fps (8 for section zoom, 10 for per-line checks) is what made texture changes and hits visible.
- **Vocabulary kept, not flattened.** On a vogue piece the per-line check, seeing only a few seconds, rewrote "the front row catwalks forward" as "struts forward". It now sees the style and the map's named elements and corrects only clear errors; the re-render named catwalks, duckwalks and face frames.
- **Definitions respected.** On a contemporary solo the check overruled the writer's "demi-pointe" with "en pointe"; the map had seen soft slippers. The fix (judge substance, don't contradict the whole-piece evidence) gave "tiny spinning bourrées on the balls of her feet".

6. Evaluation

0.81 s
median, end of prompt to first sound, 33 live lines (sessions 6 and 7; range 0.61 to 1.24 s)

311/311
demo narration clips say exactly their caption (3 checks each)

0
axe WCAG 2.2 AA violations on all three pages, light and dark, 375 and 1440 px

18
choreographed pieces described, from ballet to a 20-dancer crew

- **Live verification.** In session 7 (39.9 s) Gemini Live said "Natural spin turn, rotating rightwards across the floor"; the checker flagged "it says right, but the measured travel was left", and the next line corrected course.
- **Venue page, automated screen-reader test** (mobile viewport, fake camera of Andrii's rehearsal, keyboard only): 14 of 18 describe calls returned a line (the rest chose silence), median round trip 2.3 s, narration

announced through the live region.

- **Atlas Search:** "andri walts" (misspelt) finds "Andrii's waltz (rehearsal soundtrack)", 86.1 bpm, 3/4; "tango" finds nothing. **Hybrid Search:** "rotates to the right as a couple" ranks natural turn first in both rankings.

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7. Responsibility

- Live camera frames go to Gemini and are not stored unless the studio chooses to record. Narration goes to each listener's earphones, never over the room's music.
- In rendered videos everything outside the dancers being described is blurred. Other people's choreography is never stored in the public repository; demos read public YouTube links directly.
- Honesty about uncertainty: a term only when every part of its definition is visible; style influences named only with several distinct cues; unmeasured claims never checked as if measured; when unsure, describe the body rather than name a figure.

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8. Limitations and next

Limitations: no test with blind listeners yet; live tracking follows one dancer (groups are described from recordings); the venue page's lighter model repeats lines; live figure names are limited to those in the routine's rehearsal script; Robotics-ER target checking is planned, not built.

Next: Unseen Steps is open source under the MIT licence so any studio can run it. Next come testing with Gerard Canal's blind ballroom dancers and at real recitals, a show-notes editor in the browser, and live tracking for whole groups.